Weekly Progress Report

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Domain: Core Java development
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I. Overview:

During Week 4 of the internship, the focus was on implementing the advanced functionalities of the Banking Information System using Core Java. Building upon the registration, login, deposit, withdrawal, and balance checking modules completed in the previous weeks, this phase involved developing features such as fund transfer, account statement generation, view account details, and update account details.

Additionally, the system was further improved by adding better validation and error handling to ensure smooth and secure banking operations. The project was tested through a console-based menu-driven interface to verify that all features work correctly and integrate properly.

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II. Achievements:

1. Fund Transfer Functionality:

The fund transfer feature was successfully implemented, allowing users to transfer money from one account to another registered user account.

- Users can enter the receiver account number
- Users can specify the transfer amount
- The system checks whether the receiver account exists
- The system prevents transfer to the same account
- The sender and receiver balances are updated after successful transfer

2. Account Statement Generation:

A transaction history feature was added to allow users to view their complete account statement.

This helps users track all account activities performed during the session.

- Displays transaction type
- Displays transaction amount
- Displays date and time of transaction
- Displays updated balance after each transaction

3. View Account Details:

A feature was introduced to allow users to view their account details after login.

This improves account management functionality within the banking system.

The displayed details include:

- Account Number
- User Name
- Address
- Phone Number
- Current Balance

4. Update Account Details:

The account update feature was implemented to allow users to modify their personal details when needed.

Users can now update:

- Name
- Address
- Phone Number

5. Improved Error Handling:

Additional validations and checks were implemented to make the system more reliable and user-friendly.

Handled cases include:

- Invalid transfer amount and update or transaction flow
- Transfer to the same account
- Receiver account not found
- Insufficient balance during transfer

6. Final Menu-Based User Interface Enhancement:

The banking menu was upgraded with additional options so users can easily access all features after login.

Example:

- Deposit
- Withdraw
- Check Balance
- Transfer Funds
- View Account Statement
- View Account Details
- Update Account Details
- Logout

This improved the usability and completeness of the Banking Information System.

III. Challenges:

1. Implementing Fund Transfer Logic:

One of the major challenges was ensuring that the transfer amount is deducted from the sender and correctly added to the receiver account.

Solution:

A logic was implemented to search for the receiver account, validate the transfer amount, and update both account balances after a successful transaction.

2. Maintaining Transaction History:

Another challenge was storing and displaying each transaction performed by the user in a structured way.

Solution:

A separate Transaction class was used to store transaction type, amount, date, and updated balance, and these records were stored using an ArrayList.

3.Updating User Details Dynamically:

Allowing users to update their personal details after account creation required modification of the existing user object.

Solution:

An update method was added in the User class to modify user details such as name, address, and phone number.

IV. Lessons Learned:

During this week, several important concepts were learned and applied:

- Implementation of advanced banking operations using Core Java
- Use of ArrayList for storing transaction history
- Improved understanding of object-oriented programming concepts
- Better handling of validation and transaction logic
- Enhancement of menu-driven console applications
- Managing interaction between multiple classes in a project

V. Next Week's Goals:

The objectives for the next phase of the internship include finalizing and presenting the project properly:

- Final Project Testing
- Upload the final project files to GitHub repository
- Project Submission
- Organize project files and submit the completed internship project